

Macroeconomics I (ECO387C) Mock Midterm: Solutions

Fall 2026

Question 1

(a)

Spending on consumption and new bonds equals the endowment plus the payoff on bonds bought last period at last period's rate:

$$c_t + a_{t+1} = R_{t-1}a_t + y_t$$

(b)

Define wealth $b_t := R_{t-1}a_t$. Then the budget is $c_t + a_{t+1} = b_t + y_t$.

$$\text{state} = (b, y)$$

b is what the household brings into the period. y is a state for three reasons. It is current income. The chain is persistent, so y changes the probabilities of y' . And all households have the same y , so it is the aggregate state that determines today's bond price $R(y)$.

(c)

Let $\pi(y'|y)$ be the transition probabilities: $\pi(y_h|y_h) = p_h$, $\pi(y_l|y_h) = 1 - p_h$, $\pi(y_l|y_l) = p_l$, $\pi(y_h|y_l) = 1 - p_l$. Next period's wealth is $b' = R(y)a'$.

$$V(b, y) = \max_{a'} \left\{ u(b + y - a') + \beta \sum_{y' \in \{y_h, y_l\}} \pi(y'|y) V(R(y)a', y') \right\}, \quad u(c) = \frac{c^{1-\gamma}}{1-\gamma}.$$

Written out by state:

$$\begin{aligned} V(b, y_h) &= \max_{a'} \left\{ u(b + y_h - a') + \beta [p_h V(R(y_h)a', y_h) + (1 - p_h) V(R(y_h)a', y_l)] \right\}, \\ V(b, y_l) &= \max_{a'} \left\{ u(b + y_l - a') + \beta [(1 - p_l) V(R(y_l)a', y_h) + p_l V(R(y_l)a', y_l)] \right\}. \end{aligned}$$

(d)

FOC. Differentiate the right side with respect to a' . The derivative of $u(b + y - a')$ is $-u'(c)$. The derivative of $V(R(y)a', y')$ is $V_b(b', y') \cdot R(y)$.

$$-u'(c) + \beta \sum_{y'} \pi(y'|y) V_b(b', y') R(y) = 0$$

$$u'(c) = \beta R(y) \sum_{y'} \pi(y'|y) V_b(b', y'). \quad (1)$$

Envelope. Let $a' = g(b, y)$ be the policy and $c = b + y - g(b, y)$. Then

$$V(b, y) = u(b + y - g(b, y)) + \beta \sum_{y'} \pi(y'|y) V(R(y)g(b, y), y').$$

Differentiate with respect to b :

$$\begin{aligned} V_b(b, y) &= u'(c) \left(1 - \frac{\partial g}{\partial b}\right) + \beta \sum_{y'} \pi(y'|y) V_b(b', y') R(y) \frac{\partial g}{\partial b} \\ &= u'(c) + \frac{\partial g}{\partial b} \left[-u'(c) + \beta R(y) \sum_{y'} \pi(y'|y) V_b(b', y') \right]. \end{aligned}$$

The bracket is zero by (1), so

$$V_b(b, y) = u'(c). \quad (2)$$

Euler equation. Move (2) one period forward, $V_b(b', y') = u'(c')$, and substitute into (1):

$$u'(c) = \beta R(y) \sum_{y'} \pi(y'|y) u'(c').$$

$$\boxed{c_t^{-\gamma} = \beta R_t E_t [c_{t+1}^{-\gamma}]}$$

(e)

Market clearing (mass one of households):

$$\text{goods: } \int c_t = y_t, \quad \text{bonds: } \int a_{t+1} = 0.$$

All households are identical and face the same R_t , so they choose the same a_{t+1} . Bond clearing then gives $a_{t+1} = 0$ for everyone, and $a_t = 0$ as well. The budget becomes

$$c_t + 0 = R_{t-1} \cdot 0 + y_t \implies c_t = y_t.$$

Substitute $c_t = y_t$ and $c_{t+1} = y_{t+1}$ into the Euler equation:

$$u'(y) = \beta R(y) E[u'(y')|y]$$

$$\boxed{R(y) = \frac{u'(y)}{\beta E[u'(y')|y]} = \frac{y^{-\gamma}}{\beta E[(y')^{-\gamma}|y]}}$$

(f)

With $\gamma = 2$: $u'(y) = y^{-2}$, so $u'(2) = \frac{1}{4}$ and $u'(1) = 1$.

High state.

$$E[u'(y')|y_h] = p_h \cdot \frac{1}{4} + (1 - p_h) \cdot 1 = \frac{p_h}{4} + \frac{4 - 4p_h}{4} = \frac{4 - 3p_h}{4}$$

$$R_h = \frac{1/4}{\beta \cdot \frac{4 - 3p_h}{4}} = \frac{1}{4} \cdot \frac{4}{\beta(4 - 3p_h)} = \frac{1}{\beta(4 - 3p_h)}$$

Low state.

$$E[u'(y')|y_l] = p_l \cdot 1 + (1 - p_l) \cdot \frac{1}{4} = \frac{4p_l}{4} + \frac{1 - p_l}{4} = \frac{1 + 3p_l}{4}$$

$$R_l = \frac{1}{\beta \cdot \frac{1 + 3p_l}{4}} = \frac{4}{\beta(1 + 3p_l)}$$

$$\boxed{R_h = \frac{1}{\beta(4 - 3p_h)}, \quad R_l = \frac{4}{\beta(1 + 3p_l)}}$$

Numbers. $\beta = 0.95$, $p_h = 0.8$, $p_l = 0.6$:

$$\begin{aligned} 4 - 3p_h &= 4 - 2.4 = 1.6, & R_h &= \frac{1}{0.95 \times 1.6} = \frac{1}{1.52} = 0.658, \\ 1 + 3p_l &= 1 + 1.8 = 2.8, & R_l &= \frac{4}{0.95 \times 2.8} = \frac{4}{2.66} = 1.504, \\ & & \frac{1}{\beta} &= \frac{1}{0.95} = 1.053. \end{aligned}$$

So $R_h < 1/\beta < R_l$. In the high state income is expected to fall, so households want to save. Nobody can be on the other side of that trade, so R falls until they are willing to eat their endowment. In the low state income is expected to rise, households want to borrow, and R rises.

(g)

Why the rates are further from $1/\beta$. The rate has to offset the gap between marginal utility today and expected marginal utility tomorrow. The gap between the two states is

$$\gamma = 1 : \frac{u'(1)}{u'(2)} = \frac{1}{1/2} = 2, \quad \gamma = 2 : \frac{u'(1)}{u'(2)} = \frac{1}{1/4} = 4.$$

With $\gamma = 2$ the household dislikes uneven consumption more (its IES $1/\gamma$ is lower), so the desire to smooth is stronger. The rate has to move further to make households accept $c = y$.

iid case. If $p_h = 1 - p_l$, then $\pi(y_h|y_h) = p_h$ and $\pi(y_h|y_l) = 1 - p_l = p_h$. Next period's distribution is the same from either state, so

$$E[u'(y')|y_h] = E[u'(y')|y_l] = p_h u'(2) + (1 - p_h) u'(1) =: m.$$

Then

$$\frac{R_l}{R_h} = \frac{u'(1)/(\beta m)}{u'(2)/(\beta m)} = \frac{u'(1)}{u'(2)} = \frac{1}{1/4} = 4.$$

$$\boxed{\frac{R_l}{R_h} = 4}$$

This does not depend on p_h , p_l or β .

Question 2

(a)

Household problem. Taking $\{r_t\}_{t=0}^{\infty}$ as given, type $e \in \{A, B\}$ solves

$$\max_{\{c_t^e, b_t^e\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \ln c_t^e$$

subject to

$$c_t^e + b_t^e = y_t^e + (1 + r_{t-1})b_{t-1}^e \quad \forall t, \quad b_{-1}^e = 0, \quad \lim_{T \rightarrow \infty} \frac{b_T^e}{\prod_{\tau=0}^{T-1} (1 + r_{\tau})} \geq 0.$$

Equilibrium. A sequential competitive equilibrium is a sequence of interest rates $\{r_t\}_{t=0}^{\infty}$ and allocations $\{c_t^A, c_t^B, b_t^A, b_t^B\}_{t=0}^{\infty}$ such that

1. given $\{r_t\}$, $\{c_t^e, b_t^e\}$ solves type e 's problem for $e = A, B$,
2. the goods market clears: $c_t^A + c_t^B = y_t^A + y_t^B$ for all t ,
3. the bond market clears: $b_t^A + b_t^B = 0$ for all t .

(b)

Euler equation. Lagrangian:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \lambda_t (y_t + (1 + r_{t-1})b_{t-1} - c_t - b_t) \right].$$

FOC for c_t :

$$\beta^t \left(\frac{1}{c_t} - \lambda_t \right) = 0 \quad \implies \quad \lambda_t = \frac{1}{c_t}.$$

FOC for b_t (b_t is in the date- t constraint with coefficient -1 and in the date- $(t+1)$ constraint with coefficient $1 + r_t$):

$$-\beta^t \lambda_t + \beta^{t+1} \lambda_{t+1} (1 + r_t) = 0 \quad \implies \quad \lambda_t = \beta(1 + r_t) \lambda_{t+1}.$$

Substitute $\lambda_t = 1/c_t$:

$$\boxed{\frac{1}{c_t} = \beta(1 + r_t) \frac{1}{c_{t+1}} \quad \iff \quad c_{t+1} = \beta(1 + r_t) c_t}$$

Intertemporal budget constraint. By definition $q_0 = 1$ and $q_{t+1} = \frac{q_t}{1+r_t}$, so $q_t(1 + r_{t-1}) = q_{t-1}$. Multiply the date- t budget by q_t :

$$q_t c_t + q_t b_t = q_t y_t + q_t (1 + r_{t-1}) b_{t-1} = q_t y_t + q_{t-1} b_{t-1}.$$

Write it for $t = 0, 1, \dots, T$:

$$\begin{aligned} t = 0 : \quad & q_0 c_0 + q_0 b_0 = q_0 y_0 + 0 \\ t = 1 : \quad & q_1 c_1 + q_1 b_1 = q_1 y_1 + q_0 b_0 \\ & \vdots \\ t = T : \quad & q_T c_T + q_T b_T = q_T y_T + q_{T-1} b_{T-1} \end{aligned}$$

Add them. Each $q_{t-1} b_{t-1}$ on the right cancels the $q_{t-1} b_{t-1}$ on the left of the line above, leaving

$$\sum_{t=0}^T q_t c_t + q_T b_T = \sum_{t=0}^T q_t y_t.$$

Since $q_T = 1 / \prod_{\tau=0}^{T-1} (1 + r_\tau)$, the NPG condition holds with equality at the optimum (TVC), so $q_T b_T \rightarrow 0$. Let $T \rightarrow \infty$:

$$\boxed{\sum_{t=0}^{\infty} q_t c_t^e = \sum_{t=0}^{\infty} q_t y_t^e}$$

(c)

Start from goods clearing at t . Use the Euler equation $c_t^e = \frac{c_{t+1}^e}{\beta(1+r_t)}$ for each type, then goods clearing at $t + 1$:

$$\begin{aligned} Y_t &= c_t^A + c_t^B \\ &= \frac{c_{t+1}^A}{\beta(1+r_t)} + \frac{c_{t+1}^B}{\beta(1+r_t)} \\ &= \frac{c_{t+1}^A + c_{t+1}^B}{\beta(1+r_t)} \\ &= \frac{Y_{t+1}}{\beta(1+r_t)}. \end{aligned}$$

Solve for the rate:

$$\boxed{1 + r_t = \frac{1}{\beta} \frac{Y_{t+1}}{Y_t}}$$

Aggregate endowment: $Y_0 = 1 + 1 = 2$ and $Y_t = 1 + 2 = 3$ for $t \geq 1$.

$$\begin{aligned} 1 + r_0 &= \frac{1}{\beta} \cdot \frac{Y_1}{Y_0} = \frac{1}{\beta} \cdot \frac{3}{2} = \frac{3}{2\beta}, \\ 1 + r_t &= \frac{1}{\beta} \cdot \frac{3}{3} = \frac{1}{\beta} \quad (t \geq 1). \end{aligned}$$

Prices:

$$q_0 = 1,$$

$$\begin{aligned}
q_1 &= \frac{q_0}{1+r_0} = \frac{2\beta}{3}, \\
q_2 &= \frac{q_1}{1+r_1} = \frac{2\beta}{3} \cdot \beta = \frac{2}{3}\beta^2, \\
q_t &= \frac{2}{3}\beta^t \quad (t \geq 1).
\end{aligned}$$

$$1 + r_0 = \frac{3}{2\beta}, \quad 1 + r_t = \frac{1}{\beta} \quad (t \geq 1), \quad q_0 = 1, \quad q_t = \frac{2}{3}\beta^t \quad (t \geq 1)$$

r_0 is high because total goods rise from 2 to 3. Both types would like to borrow against the higher future, but bonds are in zero net supply, so the rate rises until net borrowing is zero.

(d)

Shape of consumption. From the Euler equation:

$$\begin{aligned}
c_1 &= \beta(1+r_0)c_0 = \beta \cdot \frac{3}{2\beta} c_0 = \frac{3}{2}c_0, \\
c_{t+1} &= \beta \cdot \frac{1}{\beta} c_t = c_t \quad (t \geq 1).
\end{aligned}$$

So each type consumes c_0^e at $t = 0$ and $\frac{3}{2}c_0^e$ at every $t \geq 1$.

Left side of the IBC.

$$\begin{aligned}
\sum_{t=0}^{\infty} q_t c_t &= 1 \cdot c_0 + \sum_{t=1}^{\infty} \frac{2}{3}\beta^t \cdot \frac{3}{2}c_0 \\
&= c_0 + c_0 \sum_{t=1}^{\infty} \beta^t \\
&= c_0 \sum_{t=0}^{\infty} \beta^t = \frac{c_0}{1-\beta}.
\end{aligned}$$

Right side of the IBC. Using $\sum_{t=1}^{\infty} \beta^t = \frac{\beta}{1-\beta}$:

$$\begin{aligned}
\text{Type A : } \sum_t q_t y_t^A &= 1 \cdot 1 + \sum_{t=1}^{\infty} \frac{2}{3}\beta^t \cdot 1 = 1 + \frac{2}{3} \cdot \frac{\beta}{1-\beta}, \\
\text{Type B : } \sum_t q_t y_t^B &= 1 \cdot 1 + \sum_{t=1}^{\infty} \frac{2}{3}\beta^t \cdot 2 = 1 + \frac{4}{3} \cdot \frac{\beta}{1-\beta}.
\end{aligned}$$

Solve. Set $\frac{c_0}{1-\beta}$ equal to the right side and multiply by $1-\beta$:

$$\begin{aligned}
c_0^A &= (1-\beta) \left(1 + \frac{2}{3} \cdot \frac{\beta}{1-\beta} \right) = (1-\beta) + \frac{2}{3}\beta = 1 - \frac{\beta}{3}, \\
c_0^B &= (1-\beta) \left(1 + \frac{4}{3} \cdot \frac{\beta}{1-\beta} \right) = (1-\beta) + \frac{4}{3}\beta = 1 + \frac{\beta}{3}.
\end{aligned}$$

Then for $t \geq 1$:

$$c_t^A = \frac{3}{2} \left(1 - \frac{\beta}{3}\right) = \frac{3}{2} - \frac{\beta}{2}, \quad c_t^B = \frac{3}{2} \left(1 + \frac{\beta}{3}\right) = \frac{3}{2} + \frac{\beta}{2}.$$

$$\boxed{c_0^A = 1 - \frac{\beta}{3}, \quad c_t^A = \frac{3}{2} - \frac{\beta}{2}; \quad c_0^B = 1 + \frac{\beta}{3}, \quad c_t^B = \frac{3}{2} + \frac{\beta}{2} \quad (t \geq 1)}$$

Check goods clearing.

$$t = 0 : \quad c_0^A + c_0^B = 1 - \frac{\beta}{3} + 1 + \frac{\beta}{3} = 2 = Y_0$$

$$t \geq 1 : \quad c_t^A + c_t^B = \frac{3}{2} - \frac{\beta}{2} + \frac{3}{2} + \frac{\beta}{2} = 3 = Y_t$$

For example, with $\beta = 0.9$: $c^A = (0.70, 1.05, 1.05, \dots)$ and $c^B = (1.30, 1.95, 1.95, \dots)$.

(e)

Date 0.

$$b_0^A = y_0^A - c_0^A = 1 - \left(1 - \frac{\beta}{3}\right) = \frac{\beta}{3}, \quad b_0^B = -b_0^A = -\frac{\beta}{3}.$$

B borrows and A lends. B 's endowment rises from 1 to 2 at $t = 1$, so B borrows against that future income to consume more today.

Date 1.

$$\begin{aligned} b_1^A &= y_1^A + (1 + r_0)b_0^A - c_1^A \\ &= 1 + \frac{3}{2\beta} \cdot \frac{\beta}{3} - \left(\frac{3}{2} - \frac{\beta}{2}\right) \\ &= 1 + \frac{1}{2} - \frac{3}{2} + \frac{\beta}{2} = \frac{\beta}{2}. \end{aligned}$$

Date $t \geq 2$. Suppose $b_{t-1}^A = \frac{\beta}{2}$. Then

$$\begin{aligned} b_t^A &= y_t^A + (1 + r_{t-1})b_{t-1}^A - c_t^A \\ &= 1 + \frac{1}{\beta} \cdot \frac{\beta}{2} - \left(\frac{3}{2} - \frac{\beta}{2}\right) \\ &= 1 + \frac{1}{2} - \frac{3}{2} + \frac{\beta}{2} = \frac{\beta}{2}. \end{aligned}$$

$$\boxed{b_0^A = \frac{\beta}{3}, \quad b_t^A = \frac{\beta}{2} \quad (t \geq 1), \quad b_t^B = -b_t^A}$$

NPG check.

$$\frac{b_T^A}{\prod_{\tau=0}^{T-1} (1 + r_\tau)} = q_T b_T^A = \frac{2}{3} \beta^T \cdot \frac{\beta}{2} = \frac{\beta^{T+1}}{3} \longrightarrow 0 \quad (T \rightarrow \infty).$$

The limit is $0 \geq 0$, so the NPG condition holds (with equality).

(f)

With log utility, $p_t = \beta^t \frac{Y_0}{Y_t}$.

$$\begin{aligned} t = 0 : \quad p_0 &= \beta^0 \cdot \frac{2}{2} = 1 = q_0, \\ t \geq 1 : \quad p_t &= \beta^t \cdot \frac{2}{3} = \frac{2}{3} \beta^t = q_t. \end{aligned}$$

$$\boxed{p_t = q_t \text{ for all } t}$$

In the sequential economy a household can move goods between any two dates by rolling over one-period bonds, at the relative price q_t . So its budget set is the same as with Arrow-Debreu date-0 trading at prices $p_t = q_t$, and it makes the same choices.

Question 3

(a)

From the resource constraint, $c = Ak^\alpha - k'$.

$$F(k, k') = \ln(Ak^\alpha - k').$$

$k' \geq 0$ and $c \geq 0$, which means $k' \leq Ak^\alpha$:

$$\Gamma(k) = [0, Ak^\alpha].$$

$$\boxed{V(k) = \max_{k' \in [0, Ak^\alpha]} \{ \ln(Ak^\alpha - k') + \beta V(k') \}}$$

The state is k and the control is k' .

(b)

Define $(Tf)(k) = \max_{k' \in \Gamma(k)} \{ \ln(Ak^\alpha - k') + \beta f(k') \}$.

Blackwell's sufficient conditions. If T satisfies

1. monotonicity: $f(k) \leq g(k)$ for all k implies $(Tf)(k) \leq (Tg)(k)$ for all k ,
2. discounting: there is $\beta \in [0, 1)$ such that $T(f+c)(k) \leq (Tf)(k) + \beta c$ for every constant $c \geq 0$ and all k ,

then T is a contraction with modulus β .

Monotonicity. Let $f \leq g$. Since $\beta > 0$, for every $k' \in \Gamma(k)$,

$$\ln(Ak^\alpha - k') + \beta f(k') \leq \ln(Ak^\alpha - k') + \beta g(k').$$

Take the max over the same set $\Gamma(k)$ on both sides:

$$(Tf)(k) = \max_{k' \in \Gamma(k)} \{\ln(Ak^\alpha - k') + \beta f(k')\} \leq \max_{k' \in \Gamma(k)} \{\ln(Ak^\alpha - k') + \beta g(k')\} = (Tg)(k).$$

Discounting. For a constant $c \geq 0$,

$$\begin{aligned} T(f+c)(k) &= \max_{k' \in \Gamma(k)} \{\ln(Ak^\alpha - k') + \beta(f(k') + c)\} \\ &= \max_{k' \in \Gamma(k)} \{\ln(Ak^\alpha - k') + \beta f(k')\} + \beta c \\ &= (Tf)(k) + \beta c. \end{aligned}$$

βc comes out of the max because it does not depend on k' . With $\beta < 1$ this is the discounting condition.

Banach. T is a contraction. On a complete metric space (bounded continuous functions with $d(f, g) = \sup_k |f(k) - g(k)|$), Banach's theorem gives

1. a unique fixed point $V = TV$, which is the value function,
2. convergence of value function iteration from any V_0 : $d(T^n V_0, V) \leq \beta d(T^{n-1} V_0, V) \leq \dots \leq \beta^n d(V_0, V) \rightarrow 0$.

Log utility. $\ln c \rightarrow -\infty$ as $c \rightarrow 0$, so F is unbounded and V is not a bounded function. The theorem does not apply directly on the space of bounded functions. One has to restrict k to a compact set bounded away from zero, or use a different norm.

(c)

Iteration 1. With $V_0 \equiv 0$:

$$V_1(k) = \max_{0 \leq k' \leq Ak^\alpha} \ln(Ak^\alpha - k').$$

$\ln(Ak^\alpha - k')$ is decreasing in k' , so the max is at $k' = 0$:

$$V_1(k) = \ln(Ak^\alpha) = \ln A + \alpha \ln k.$$

$$\boxed{V_1(k) = \ln A + \alpha \ln k, \quad g_1(k) = 0}$$

Iteration 2.

$$V_2(k) = \max_{k'} \{\ln(Ak^\alpha - k') + \beta(\ln A + \alpha \ln k')\}.$$

FOC with respect to k' :

$$-\frac{1}{Ak^\alpha - k'} + \frac{\alpha\beta}{k'} = 0 \implies \frac{1}{Ak^\alpha - k'} = \frac{\alpha\beta}{k'}.$$

Cross-multiply and solve:

$$k' = \alpha\beta(Ak^\alpha - k')$$

$$\begin{aligned}
k' &= \alpha\beta Ak^\alpha - \alpha\beta k' \\
k' + \alpha\beta k' &= \alpha\beta Ak^\alpha \\
k'(1 + \alpha\beta) &= \alpha\beta Ak^\alpha
\end{aligned}$$

$$g_2(k) = \frac{\alpha\beta}{1 + \alpha\beta} Ak^\alpha$$

Interpretation. V_n is the value of the problem with n periods left. g_2 is the optimal choice with two periods left: save some, knowing that everything will be eaten in the last period. It is the finite-horizon rule $k_T = \frac{\alpha\beta}{1+\alpha\beta} Ak_{T-1}^\alpha$. With $\alpha = 0.3$, $\beta = 0.95$ the saving rate is $\frac{0.285}{1.285} = 0.222$.

(d)

Derivatives of F . With $c = Ak^\alpha - k'$:

$$F_1(k, k') = \frac{\alpha Ak^{\alpha-1}}{Ak^\alpha - k'}, \quad F_2(k, k') = -\frac{1}{Ak^\alpha - k'}.$$

FOC. $F_2(k, k') + \beta V'(k') = 0$:

$$\frac{1}{Ak^\alpha - k'} = \beta V'(k'). \quad (3)$$

Envelope. With $k' = g(k)$, $V(k) = F(k, g(k)) + \beta V(g(k))$. Differentiate:

$$\begin{aligned}
V'(k) &= F_1(k, g(k)) + F_2(k, g(k))g'(k) + \beta V'(g(k))g'(k) \\
&= F_1(k, g(k)) + g'(k)[F_2(k, g(k)) + \beta V'(g(k))] \\
&= F_1(k, g(k)) \quad (\text{bracket} = 0 \text{ by (3)}) \\
&= \frac{\alpha Ak^{\alpha-1}}{Ak^\alpha - g(k)}.
\end{aligned}$$

Move forward one period:

$$V'(g(k)) = \frac{\alpha A g(k)^{\alpha-1}}{A g(k)^\alpha - g(g(k))}.$$

Functional Euler equation. Substitute into (3):

$$\frac{1}{Ak^\alpha - g(k)} = \beta \frac{\alpha A g(k)^{\alpha-1}}{A g(k)^\alpha - g(g(k))}.$$

Guess $g(k) = sAk^\alpha$. Then $g(g(k)) = sA g(k)^\alpha$.

$$\text{Left side: } \frac{1}{Ak^\alpha - sAk^\alpha} = \frac{1}{(1-s)Ak^\alpha}.$$

$$\text{Right side: } \beta \frac{\alpha A g(k)^{\alpha-1}}{A g(k)^\alpha - sA g(k)^\alpha} = \beta \frac{\alpha A g(k)^{\alpha-1}}{(1-s)A g(k)^\alpha} = \frac{\alpha\beta}{(1-s)g(k)} = \frac{\alpha\beta}{(1-s)sAk^\alpha}.$$

Set them equal:

$$\frac{1}{(1-s)Ak^\alpha} = \frac{\alpha\beta}{(1-s)sAk^\alpha}.$$

Multiply both sides by $(1-s)Ak^\alpha$:

$$1 = \frac{\alpha\beta}{s} \implies s = \alpha\beta.$$

k cancels, so the guess works for every k .

$$\boxed{s = \alpha\beta, \quad g(k) = \alpha\beta Ak^\alpha, \quad c = (1 - \alpha\beta)Ak^\alpha}$$

(e)

At the steady state $k' = k = k^*$:

$$\begin{aligned} k^* &= \alpha\beta A(k^*)^\alpha \\ (k^*)^{1-\alpha} &= \alpha\beta A \\ k^* &= (\alpha\beta A)^{\frac{1}{1-\alpha}}. \end{aligned}$$

Numbers: $\alpha\beta A = 0.3 \times 0.95 \times 1 = 0.285$ and $\frac{1}{1-\alpha} = \frac{1}{0.7}$.

$$\boxed{k^* = 0.285^{1/0.7} = 0.166}$$

(f)

Take logs of the policy:

$$\ln k_{t+1} = \ln(\alpha\beta A) + \alpha \ln k_t.$$

At the steady state:

$$\ln k^* = \ln(\alpha\beta A) + \alpha \ln k^*.$$

Subtract the second equation from the first:

$$\ln k_{t+1} - \ln k^* = \alpha(\ln k_t - \ln k^*)$$

$$\boxed{\hat{k}_{t+1} = \alpha \hat{k}_t}$$

This is exact, because the policy is linear in logs.

$k_0 = k^*/2$ gives $\hat{k}_0 = \ln \frac{1}{2} = -0.693$. So

$$\hat{k}_1 = 0.3 \times (-0.693) = -0.208.$$

Iterating, $\hat{k}_n = \alpha^n \hat{k}_0$. Half the gap is closed when $\alpha^n = \frac{1}{2}$:

$$n = \frac{\ln 0.5}{\ln 0.3} = \frac{-0.693}{-1.204} = 0.58 \text{ periods.}$$

More than half the gap is closed in the first period.

Question 4

(a)

Write the budget as $c_t + \frac{a_{t+1}}{1+r} = a_t + y_t$. Lagrangian:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[u(c_t) + \lambda_t \left(a_t + y_t - c_t - \frac{a_{t+1}}{1+r} \right) \right].$$

FOC for c_t : $u'(c_t) = \lambda_t$. FOC for a_{t+1} :

$$-\beta^t \frac{\lambda_t}{1+r} + \beta^{t+1} E_t \lambda_{t+1} = 0 \implies \lambda_t = \beta(1+r) E_t \lambda_{t+1}.$$

So $u'(c_t) = \beta(1+r) E_t u'(c_{t+1})$. With $\beta(1+r) = 1$:

$$u'(c_t) = E_t u'(c_{t+1}).$$

With quadratic utility $u'(c) = b_1 - b_2 c$:

$$b_1 - b_2 c_t = E_t [b_1 - b_2 c_{t+1}]$$

$$b_1 - b_2 c_t = b_1 - b_2 E_t c_{t+1}$$

$$c_t = E_t c_{t+1}.$$

$$\boxed{c_t = E_t c_{t+1}}$$

(b)

From the budget:

$$c_t = a_t + y_t - \frac{a_{t+1}}{1+r}.$$

Write the same equation at $t+1$ and solve for a_{t+1} :

$$a_{t+1} = c_{t+1} - y_{t+1} + \frac{a_{t+2}}{1+r}.$$

Substitute:

$$\begin{aligned} c_t &= a_t + y_t - \frac{1}{1+r} \left(c_{t+1} - y_{t+1} + \frac{a_{t+2}}{1+r} \right) \\ c_t + \frac{c_{t+1}}{1+r} &= a_t + y_t + \frac{y_{t+1}}{1+r} - \frac{a_{t+2}}{(1+r)^2}. \end{aligned}$$

Repeating J times gives

$$\sum_{j=0}^J \frac{c_{t+j}}{(1+r)^j} = a_t + \sum_{j=0}^J \frac{y_{t+j}}{(1+r)^j} - \frac{a_{t+J+1}}{(1+r)^{J+1}}.$$

Take E_t and let $J \rightarrow \infty$. The last term goes to zero by the NPG condition and the TVC:

$$\sum_{j=0}^{\infty} \frac{E_t c_{t+j}}{(1+r)^j} = a_t + \sum_{j=0}^{\infty} \frac{E_t y_{t+j}}{(1+r)^j}.$$

By (a) and the law of iterated expectations, $E_t c_{t+2} = E_t[E_{t+1} c_{t+2}] = E_t c_{t+1} = c_t$, and in general $E_t c_{t+j} = c_t$. The left side is

$$c_t \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j = c_t \cdot \frac{1}{1 - \frac{1}{1+r}} = c_t \cdot \frac{1+r}{r}.$$

So

$$c_t \cdot \frac{1+r}{r} = a_t + \sum_{j=0}^{\infty} \frac{E_t y_{t+j}}{(1+r)^j}$$

$$\boxed{c_t = \frac{r}{1+r} \left[a_t + \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j E_t y_{t+j} \right]}$$

(c)

By (a), $c_{t-1} = E_{t-1} c_t$, so

$$\Delta c_t = c_t - c_{t-1} = c_t - E_{t-1} c_t.$$

Apply (b) at t and take E_{t-1} of it. a_t was chosen at $t-1$, so $E_{t-1} a_t = a_t$ and it cancels:

$$\Delta c_t = \frac{r}{1+r} \sum_{j=0}^{\infty} \frac{E_t y_{t+j} - E_{t-1} y_{t+j}}{(1+r)^j}.$$

Revision in expected income. Iterating the AR(1):

$$E_t y_{t+j} - \bar{y} = \rho^j (y_t - \bar{y}), \quad E_{t-1} y_{t+j} - \bar{y} = \rho^{j+1} (y_{t-1} - \bar{y}).$$

Subtract:

$$E_t y_{t+j} - E_{t-1} y_{t+j} = \rho^j [(y_t - \bar{y}) - \rho(y_{t-1} - \bar{y})] = \rho^j \varepsilon_t.$$

Sum. $\frac{\rho}{1+r} < 1$, so the geometric series converges:

$$\begin{aligned} \Delta c_t &= \frac{r}{1+r} \varepsilon_t \sum_{j=0}^{\infty} \left(\frac{\rho}{1+r} \right)^j \\ &= \frac{r}{1+r} \cdot \frac{1}{1 - \frac{\rho}{1+r}} \varepsilon_t \\ &= \frac{r}{1+r} \cdot \frac{1+r}{1+r-\rho} \varepsilon_t \end{aligned}$$

$$\boxed{\Delta c_t = \frac{r}{1+r-\rho} \varepsilon_t}$$

Numbers, $r = 0.04$:

$$\begin{aligned}\rho = 0 : \quad & \frac{0.04}{1.04} = 0.038, \\ \rho = 0.9 : \quad & \frac{0.04}{1.04 - 0.9} = \frac{0.04}{0.14} = 0.286, \\ \rho = 1 : \quad & \frac{0.04}{1.04 - 1} = \frac{0.04}{0.04} = 1.\end{aligned}$$

The more persistent the shock, the more consumption responds. A one-time shock raises consumption only by its annuity value (about r). A permanent shock raises consumption one for one.

(d)

The model predicts a coefficient of zero. $\Delta c_{t+1} = c_{t+1} - E_t c_{t+1}$ is a rational-expectations forecast error, so it is uncorrelated with anything in the time- t information set:

$$E_t[x_t \Delta c_{t+1}] = x_t E_t[\Delta c_{t+1}] = 0, \quad \text{slope} = \frac{\text{Cov}(\Delta c_{t+1}, x_t)}{\text{Var}(x_t)} = 0.$$

So $H_0 : \gamma = 0$ in $\Delta c_{t+1} = \gamma x_t + \text{error}$. Johnson, Parker and Souleles (2006) used the 2001 tax rebates (\$300–600), whose timing was random and known in advance. The PIH says spending should not respond when the check arrives. They found households spent about 20–30% of the rebate in the quarter it arrived, which rejects the PIH (excess sensitivity).

(e)

Unconstrained rule. Income is iid, so $E_t y_{t+j} = \bar{y}$ for $j \geq 1$. From (b):

$$\begin{aligned}c_t &= \frac{r}{1+r} \left[a_t + y_t + \sum_{j=1}^{\infty} \frac{\bar{y}}{(1+r)^j} \right] \\ &= \frac{r}{1+r} \left[a_t + y_t + \frac{\bar{y}}{r} \right] \\ &= \frac{r}{1+r} (a_t + y_t) + \frac{\bar{y}}{1+r}.\end{aligned}$$

using $\sum_{j=1}^{\infty} (1+r)^{-j} = \frac{1/(1+r)}{1-1/(1+r)} = \frac{1}{r}$.

Numbers. $a_t = 0$, $y_t = 0.5$, $\bar{y} = 1$, $r = 0.04$:

$$c_t = \frac{0.04}{1.04} (0 + 0.5) + \frac{1}{1.04} = 0.0192 + 0.9615 = 0.981.$$

Violation.

$$a_{t+1} = (1+r)(a_t + y_t - c_t) = 1.04 \times (0 + 0.5 - 0.981) = 1.04 \times (-0.481) = -0.500 < 0.$$

The rule asks the household to borrow, which $a_{t+1} \geq 0$ rules out.

Euler equation with the constraint. Add $\mu_t(a_{t+1} - 0)$ to the Lagrangian in (a). The FOC for a_{t+1} becomes

$$-\beta^t \frac{\lambda_t}{1+r} + \beta^{t+1} E_t \lambda_{t+1} + \beta^t \mu_t = 0.$$

Divide by β^t , multiply by $1+r$, and use $\lambda = u'$:

$$\boxed{u'(c_t) = \beta(1+r)E_t u'(c_{t+1}) + (1+r)\mu_t, \quad \mu_t \geq 0, \quad \mu_t a_{t+1} = 0}$$

Consumption and MPC. The constraint binds, $a_{t+1} = 0$, so $\mu_t > 0$ and $u'(c_t) > \beta(1+r)E_t u'(c_{t+1})$. From the budget:

$$0 = (1+r)(a_t + y_t - c_t) \implies c_t = a_t + y_t = 0 + 0.5 = 0.5.$$

$$\text{MPC} = \frac{\partial c_t}{\partial y_t} = 1.$$

$$\boxed{c_t = 0.5, \quad \text{MPC} = 1}$$

The unconstrained MPC out of a transitory shock would be $\frac{r}{1+r} = 0.038$.

(f)

With $\beta(1+r) = 1$ the Euler equation is $u'(c_t) = E_t u'(c_{t+1})$. Expand $u'(c_{t+1})$ to second order around c_t :

$$u'(c_{t+1}) \approx u'(c_t) + u''(c_t)(c_{t+1} - c_t) + \frac{1}{2}u'''(c_t)(c_{t+1} - c_t)^2.$$

Take E_t and use $E_t u'(c_{t+1}) = u'(c_t)$:

$$0 = u''(c_t)E_t[c_{t+1} - c_t] + \frac{1}{2}u'''(c_t)E_t[(c_{t+1} - c_t)^2].$$

Solve for $E_t[c_{t+1} - c_t]$ and divide by c_t :

$$E_t \left[\frac{c_{t+1} - c_t}{c_t} \right] = -\frac{u'''(c_t)c_t}{u''(c_t)} \cdot \frac{1}{2} E_t \left[\left(\frac{c_{t+1} - c_t}{c_t} \right)^2 \right].$$

For CRRA, $u''(c) = -\gamma c^{-\gamma-1} < 0$ and $u'''(c) = \gamma(\gamma+1)c^{-\gamma-2} > 0$, so

$$-\frac{u'''(c)c}{u''(c)} = \frac{\gamma(\gamma+1)c^{-\gamma-2}c}{\gamma c^{-\gamma-1}} = \gamma+1 > 0.$$

Expected consumption growth is positive and increases with the variance of consumption growth. For given lifetime resources, a steeper expected path means lower consumption today, so the household saves more when income risk rises. With quadratic utility $u''' = 0$ and this term is zero, which is why there was no precautionary saving in (a)–(c).

Picture: u' is convex when $u''' > 0$, so by Jensen $E_t u'(c_{t+1}) > u'(E_t c_{t+1})$. Then $u'(c_t) = E_t u'(c_{t+1}) > u'(E_t c_{t+1})$, and since u' is decreasing, $c_t < E_t c_{t+1}$.